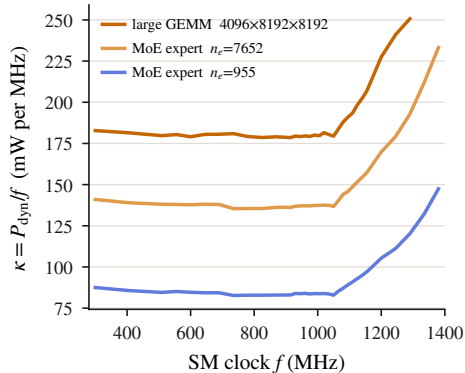
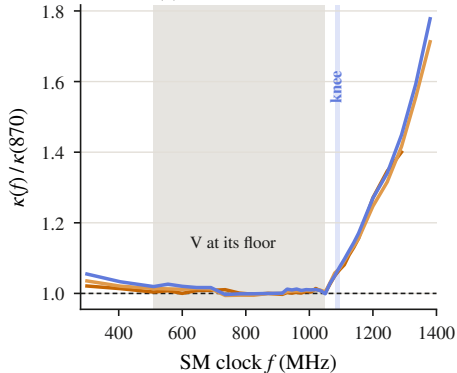


(a) as measured: levels span 2.2×



(b) normalised: one curve



$\kappa \equiv P_{\text{dyn}}/f = CV^2$: dynamic power (board power minus the idle floor sampled at that same clock) divided by frequency, which is the energy per unit of switching and therefore a direct read of V^2 . Changing the workload changes C and so the level; only the rail can change the shape. Measured over 33 clocks in one pass, timing by CUDA graph replay of 64 GEMMs so the launch gap cannot masquerade as kernel time. Flatness over 510-1035 MHz and the knee, per workload:

MoE expert: $\pm 2.0\%$, knee 1080 MHz, $\kappa(1380)/\kappa(870)=1.71$ MoE expert: $\pm 2.9\%$, knee 1095 MHz, $\kappa(1380)/\kappa(870)=1.78$ large GEMM: $\pm 1.7\%$, knee 1095 MHz (1380 throttled).